Data Science Midterm Project Report

Team: Graph Out Loud

Delayed in New York: does weather, airline or season decide it, and can a late flight catch up in the air?

Group name	Graph Out Loud
Dataset	BTS On-Time Performance, all 2013 departures from JFK, LGA and EWR (transtats.bts.gov), with NOAA hourly airport weather; packaged as github.com/hadley/nycflights13 and pypi.org/project/nycflights13
Code	github.com/Nikkumar-12345/cse558-midterm-graph-out-loud (`analysis.py` and README)

#	Name	Roll no.	Institute email	Contribution
1	Vatsal Garg	2024611	vatsal24611@iiitd.ac.in	Dataset and preprocessing (1, 2, 3)
2	Nikhil Kumar	2024382	nikhil24382@iiitd.ac.in	Exploratory data analysis and problem statements (4)
3	Shakti	2024519	shakti24519@iiitd.ac.in	In-flight delay recovery and weather impact analysis (5: P1, P2)
4	Navanshu Nagar	2024374	navanshu24374@iiitd.ac.in	Carrier performance and seasonal cancellation analysis (5: P3, P4)
5	M. Rayhankhan	2022269	rayhan22269@iiitd.ac.in	Weekly delay patterns, synthesis, limitations, code and README (5, 6)

1. Dataset

Source: U.S. Bureau of Transportation Statistics (BTS) On-Time Performance records for every domestic flight that left New York City's three airports (JFK, LaGuardia, Newark) in 2013, as released in the `nycflights13` package. One row is one scheduled flight. We join the package's `weather` table (NOAA hourly observations at the same airports, 26,115 rows $\times$ 15 columns) on airport and hour.

What is raw about it: clock times are integers (`517` is 05:17, and `2400` appears 179 times); a cancelled flight is not labelled and shows up only as a missing `dep_time`; a diverted flight shows up as a missing `arr_delay`; the weather table has a wind speed of 1,048 mph, 165 missing hours and a gust column that is 79.6% empty; delays run from -86 to $+1{,}301$ minutes.

	Raw file	After encoding
Instances (rows)	336,776	336,776
Attributes (columns)	19	167
Instances $\times$ attributes	6,398,744	56,241,592

Size check: raw, $336{,}776 \times 19 = 6{,}398{,}744 \geq 3{,}000{,}000$, so the requirement is already met. After encoding, $336{,}776 \times 167 = 56{,}241{,}592$. The 167 attributes are 28 numeric columns (12 from the flights file, with clock times in minutes; 8 joined weather values; a `gusty` flag; and 7 derived columns: `gain`, `cancelled`, `wet`, `incomplete`, `weather_missing`, `weekend`, `late15`), 4 sine/cosine columns for departure clock time and day of year, and 135 one-hot columns: `dest` (105), `carrier` (16), `weekday` (7), `season` (4) and `origin` (3). `analysis.py` builds this matrix and checks its shape.

Key features

Feature	Type	Real-world meaning
`dep_delay`	continuous	minutes the plane left the gate after schedule (negative = early)
`arr_delay`	continuous	minutes late at the destination gate
`gain`	continuous	derived: `dep_delay` $-$ `arr_delay`, minutes made up en route
`carrier`	nominal	operating airline, 16 levels
`cancelled`	binary	derived: the flight never departed
`precip`	continuous	precipitation (inches) at the origin in the scheduled hour
`wet`	binary	derived: `precip` > 0
`season`	nominal	from month: Dec–Feb winter, Mar–May spring, Jun–Aug summer, Sep–Nov autumn
`origin`, `dest`	nominal	3 NYC airports; 105 destinations

2. Data preparation

Issue found	What we did	Why, and not the alternative
8,255 missing `dep_time`, `dep_delay`	coded `cancelled` $= 1$; kept the rows	a missing departure is a cancellation. Dropping the rows would delete Test 4's response.
1,175 departed, no `arr_delay`	flagged `incomplete`; left out of arrival tests only	diverted or unrecorded arrivals. An imputed arrival delay would create a made-up gain in Test 1.
2,512 missing `tailnum`	left as is	all 2,512 are cancelled flights (checked), so no aircraft was assigned
clock times as integers, `2400`	converted to minutes after midnight, mod 1440	12:59 to 13:00 is 1 minute, not 41
wind speed 1,048 mph (1 row)	set to missing	physically impossible; a sensor error
`wind_gust` 79.6% missing	replaced by a 0/1 `gusty` flag	gusts are recorded only when there is one, so missing means none
165 weather hours absent	left join on (origin, UTC hour); 1,556 flights flagged `weather_missing`	an inner join would drop these flights from every test; only Test 2 leaves them out. The UTC hour avoids the repeated 1 a.m. hour when clocks go back (3 such rows).
27,789 arrival delays above the IQR fence (60.5 min)	kept, not capped	long delays are real. Capping would pull every mean down; the tests below are chosen to cope with the skew.
duplicates	checked full rows, (date, carrier, flight, origin, scheduled time), and (origin, hour) in weather	none found
`dest` 105, `carrier` 16 levels	one-hot; clock time and day of year as sine/cosine	codes have no order; sine/cosine keep 23:00 next to 00:00

3. What changed after preparation

Measure	Before	After
Rows	336,776	336,776 (none dropped)
Attributes	19	167
Unexplained missing `dep_delay`	8,255	0: all labelled `cancelled` (2.45%)
Unexplained missing `arr_delay`	9,430	0: 8,255 cancelled + 1,175 `incomplete`
Flights with weather	0 (separate table)	335,220; 1,556 flagged
Impossible wind values	1	0
Clock values `2400`	179	0
Duplicate rows	0	0
Arrival delays above IQR fence	27,789 (8.5%)	27,789, kept on purpose

No row was lost. The 8,255 missing departures became a response variable (`cancelled`). Arrival delay keeps its skewness of 3.7 and excess kurtosis of 29 because the long tail was kept, so each test below is checked with a rank-based or day-level version.

4. Basic questions and problem statement

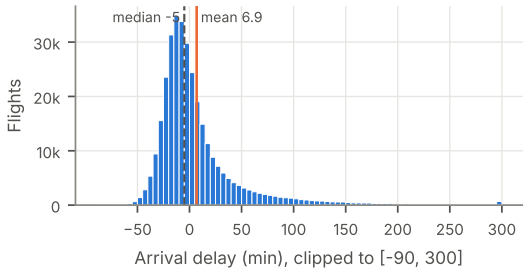


Figure 1: 57.7% of completed flights arrive early (median -5 min), but the long right tail (skewness 3.7) pulls the mean up to $+6.9$.

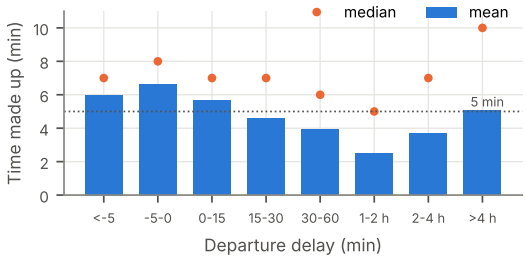


Figure 2: Flights that leave on time make up about 6 min en route; flights that leave 30 min to 2 h late make up less, not more.

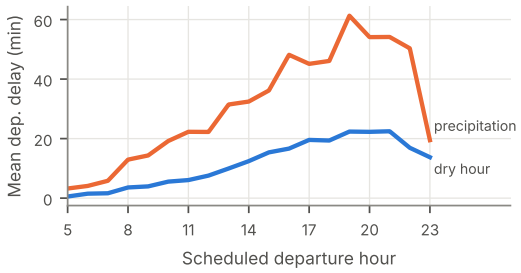


Figure 3: Delay builds through the day, and hours with precipitation sit above dry hours at every departure time.

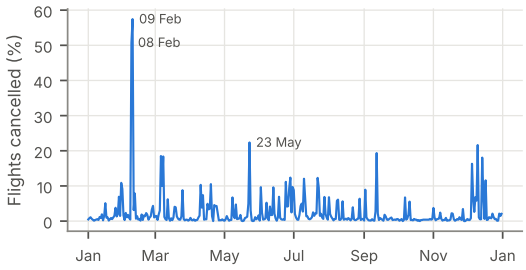


Figure 4: The median day cancels 0.73% of flights; on 8 and 9 February (a blizzard) more than half were cancelled.

Basic questions

1. Which airline has the highest average arrival delay? Frontier (F9, 21.9 min), but it has only 681 completed flights; among the 11 large carriers, AirTran (FL, 20.1 min).
2. Which month or season has the most cancellations? February (1,261 flights, 5.05%); winter by season (3.50%).
3. Are departure and arrival delays related? Yes: $r = 0.91$ over all completed flights and 0.94 over flights that left 15 or more minutes late.

Our problem statements

P1 (primary). Among flights that leave at least 15 minutes late, is the average time made up in the air more than 5 minutes?

P2. Is a flight scheduled in an hour with precipitation at its origin delayed more than 15 minutes longer, on average, than one scheduled in a dry hour?

P3. Do carriers differ in arrival delay, and by 10 minutes or more for any pair?

P4. Does the cancellation rate depend on season, and does that survive removing the worst storm days?

P5. Do weekend days have a different mean arrival delay from weekdays?

How they differ from the basic questions: the basic questions summarise one column each. P1, P2 and P4 need columns that no raw row has (`gain`, `wet` from the hourly weather join, `cancelled`), and each statement fixes a threshold for a difference that matters.

5. Statistical inferences

Rows tested: Test 1: the 72,420 completed flights with `dep_delay` ≥ 15. Test 2: the 326,993 departed flights with weather (21,086 wet, 305,907 dry). Test 3: the 325,041 completed flights of the 11 carriers with at least 1,000 of them (AS, F9, YV, HA and OO, 2,305 flights, are left out). Test 4: all 336,776 scheduled flights, summarised as 365 daily rates. Test 5: the 327,346 completed flights, summarised as 365 daily means.

How the hypotheses are set: P1 and P2 ask whether an effect is larger than a threshold, so the threshold is the boundary of H_0. P3 to P5 ask whether groups differ, so H_0 is no difference and the threshold is used to judge the confidence interval. Flights on the same day share weather and congestion, so they are not independent: the main tests use standard errors clustered by day (a t distribution with one less degree of freedom than the number of days) or take the day as the unit.

Test 1 • In-flight delay recovery (P1): do late flights make up more than 5 minutes in the air?

a) Hypotheses, written before running the test

H_0	$\mu_g \leq 5$
H_1	$\mu_g > 5$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	one-sample t-test on g (paired design), standard errors clustered by day
Smallest difference that matters	5 minutes

$g =$ `dep_delay` $-$ `arr_delay` for one flight; μ_g is its mean over flights that left at least 15 minutes late. 5 minutes is a third of the FAA's 15-minute line for a delayed flight; a smaller gain rarely changes whether a passenger arrives late.

b) Other tests we considered

Each flight gives two measurements of the same trip, so the data are paired and the test works on the differences g. With $n = 72{,}420$ the mean of g is close to normal despite its skewness (-1.3). Late flights on one day share the same congestion, so the standard error is clustered by day (365 days).

Also considered	Why we did not use it
Paired t-test, flights independent	ignores the day clustering; its standard error is 7 times smaller. Result: $t = -15.18$ (72,419), CI 3.63 to 3.95.
Wilcoxon signed-rank on $g - 5$	tests the pseudo-median, which differs from the mean when g is skewed. Result: $z = 8.08$, $p < 0.001$ in favour of > 5.
Sign test against 5	tests the median and ignores sizes: 38,125 flights gain more than 5 min and 32,510 less, $p < 0.001$. Median gain 7 min (day-bootstrap CI 6 to 7).
Welch two-sample t	treats departure and arrival delays as unrelated samples; against 0 it gives $t = 11.5$ instead of 47.5 for the paired test
z-test	needs a known σ

c) Result

Estimate	$+3.79$ min (left 64.6 min late, arrived 60.9 min late)
95% confidence interval	2.71 to 4.87 min (day-clustered)
Test statistic (d.f.)	$t = -2.21$ (364)
p-value	0.99 (one-sided)
Decision	fail to reject H_0

Late flights make up about 4 minutes on average, 6% of their 64.6-minute delay. Against zero the gain is clear ($t = 6.92$), but the whole interval lies below 5 minutes, and the reverse test (H_1: $\mu_g < 5$) gives $p = 0.014$. Cohen's $d_z = 0.18$.

The median tells a different story: the typical late flight gains 7 minutes, and the Wilcoxon and sign tests put it above 5. The mean is lower because a minority of flights lose a lot of time in the air. So the answer to P1, which is about the average, is no.

The gain is also not a catch-up. Flights that left less than 15 minutes late gain more (6.2 min), the gain does not grow with departure delay (Spearman $\rho = -0.02$), and late flights fly their route about as fast as the rest (air time 0.85 min above the route median, against 1.17 for other flights). It looks like schedule padding that every flight gets (Figure 2).

With one mean per day instead (365 days, each weighted equally) the gain is 5.86 min (CI 4.99 to 6.72; $t = 1.94$, $p = 0.027$ against 5). This is a different average: on the quarter of days with the fewest late flights they gain 10.1 min, on the quarter with the most they gain nothing (-0.1 min; Spearman $\rho = -0.46$ between the two). Weighting days equally gives the quiet days more say. On a bad day, late flights do not make up time.

Test 2 • Weather impact on departures (P2): does precipitation add more than 15 minutes of delay?

a) Hypotheses, written before running the test

H_0	$\mu_{\text{wet}} - \mu_{\text{dry}} \leq 15$
H_1	$\mu_{\text{wet}} - \mu_{\text{dry}} > 15$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	difference in means (Welch form), standard errors clustered by day
Smallest difference that matters	15 minutes

μ_{wet}, μ_{dry}: mean departure delay of flights scheduled in an hour with and without precipitation at their origin. 15 minutes is the FAA's line for a delayed flight.

b) Other tests we considered

Two independent groups, a continuous response and unknown σ call for a two-sample t-test; the spreads differ (SD 59.2 vs 38.3 min; Levene $W = 3{,}916$, $p < 0.001$), so the Welch form. Flights in one hour share its weather, and the hourly mean delay at an airport is strongly autocorrelated (lag-1 $r = 0.81$), so standard errors are clustered by day.

Also considered	Why we did not use it
Welch t-test, flights independent	ignores that all flights in a wet hour share the same weather. Result: $t = 11.0$ (22,314), CI 18.7 to 20.4.
Pooled t-test	equal variances are rejected; against 0 it gives $t = 68.8$
Mann–Whitney U	compares ranks, not minutes. $P(\text{wet} > \text{dry}) = 0.64$, $p < 0.001$; medians 5 vs -2 min.
Permutation test	shuffling single flights breaks the clustering; gives only a p-value
z-test	needs a known σ

c) Result

Estimate	$+19.6$ min (30.9 wet, 11.4 dry)
95% confidence interval	14.8 to 24.3 min (day-clustered)
Test statistic (d.f.)	$t = 1.88$ (363)
p-value	0.030 (one-sided)
Decision	reject H_0 at 0.05

Precipitation goes with about 20 more minutes of departure delay, and against zero this is clear ($t = 8.09$). That it is more than 15 minutes is borderline. The 95% interval starts at 14.8; the one-sided test at 0.05 matches the 90% interval, 15.6 to 23.5, which just clears 15. 40% of wet-hour departures are 15 or more minutes late against 21% of dry ones, and delay rises with intensity: 27.3, 30.4 and 40.9 min for up to 0.02, 0.02 to 0.1 and over 0.1 inch.

Two checks. (i) One row per airport-hour (19,326 hours) gives $+20.5$ min (CI 18.3 to 22.6; Welch $t = 5.04$, d.f. 1,328, $p < 0.001$ against 15), but neighbouring hours are still correlated. (ii) Rain may fall more often in months and hours that have more delay anyway. With month, hour and airport fixed effects (errors clustered by day) the gap is $+18.5$ min (CI 14.1 to 22.9; $t = 1.57$, $p = 0.059$), and H_0 is no longer rejected. Precipitation clearly adds delay; that it adds more than 15 minutes is not established.

Test 3 • Carrier performance (P3): do carriers differ in arrival delay?

a) Hypotheses, written before running the test

H_0	the arrival-delay distributions of the 11 carriers are identical
H_1	at least one carrier's delays tend to be larger or smaller
Significance level	$\alpha = 0.05$
Side	omnibus (upper tail of H)
Test	Kruskal–Wallis, then Dunn's test with Holm adjustment
Smallest difference that matters	10 minutes between two carriers

10 minutes is two-thirds of the FAA line, a gap a frequent traveller would notice. A rank test cannot put minutes in H_0, so the threshold is applied to simultaneous 95% intervals for the pairwise mean differences (Games–Howell).

b) Other tests we considered

Eleven independent groups, a heavily skewed response (per-carrier skewness 2.6 to 5.4) and unequal spreads (Levene $W = 207.5$, $p < 0.001$). Kruskal–Wallis compares ranks, so a few 20-hour delays cannot dominate it, and it does not need normality. Dunn's test is the matching rank-based follow-up.

Also considered	Why we did not use it
One-way ANOVA	assumes equal variances, which Levene rejects. Result: $F = 477.7$ (10, 325,030), same decision.
Welch ANOVA	handles unequal variances but compares means, which the tails drive. Result: $F = 445.5$ (10, 50,794), $\eta^2 = 0.014$.
55 pairwise t-tests	the chance of at least one false rejection would be $1 - 0.95^{55} = 94\%$

c) Result

Estimate	worst (AirTran, FL) minus best (American, AA): $+19.8$ min mean (20.1 vs 0.4); medians 5 vs -9
95% confidence interval	16.6 to 22.9 min (Games–Howell, simultaneous); median gap 13 to 16 (bootstrap)
Test statistic (d.f.)	$H = 8{,}170.2$ (10)
p-value	< 0.001
Decision	reject H_0

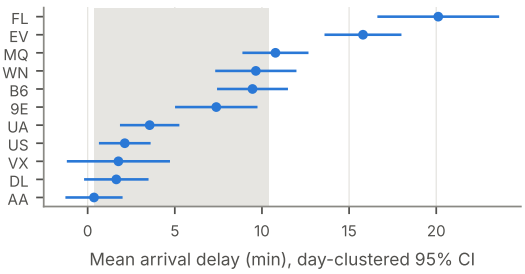


Figure 5: Mean arrival delay by carrier with day-clustered 95% CIs. The grey band spans 10 min above the best carrier; only EV and FL sit clearly outside it.

Carriers differ, but carrier explains little: $\varepsilon^2 = 0.025$, about 2.5% of the variation in ranks. Dunn–Holm finds 53 of 55 pairs different (not AA–VX or B6–WN). Only 14 pairs differ by 10 minutes or more in mean, and 13 of them involve ExpressJet (EV, 15.8 min) or AirTran (FL, 20.1); the other is MQ against AA. Using the simultaneous intervals, 10 pairs are more than 10 minutes apart for certain, 34 are less than 10 apart for certain, and 11 are undecided. By medians only 6 pairs reach 10 minutes, all involving FL.

Because flights on one day share conditions, we checked the result with the day built in. A Friedman test with days as blocks (each carrier's median delay per day, 365 days) gives $\chi^2 = 1{,}144.6$ (10), $p < 0.001$, Kendall's $W = 0.31$, so the order holds within days. For FL − AA, day-clustered errors give CI 17.1 to 22.4 and day fixed effects give $+20.1$ (CI 17.4 to 22.9). Kruskal–Wallis within each airport is significant at all three (EWR $H = 3{,}306$, 9 d.f.), so the gap is not about which airport a carrier uses.

Test 4 • Seasonal cancellation risk (P4): do cancellations depend on season?

a) Hypotheses, written before running the test

H_0	the daily cancellation rate has the same distribution in all four seasons
H_1	it differs in at least one season
Significance level	$\alpha = 0.05$
Side	omnibus (upper tail of H)
Test	Kruskal–Wallis on the 365 daily cancellation rates
Smallest difference that matters	1 percentage point between two seasons

Against a yearly rate of 2.45%, a 1-point gap is a 40% change in a traveller's risk, about 840 extra cancelled flights per season out of New York.

b) Other tests we considered

Cancellations come in bursts: the 10 worst days hold 27.2% of the year's cancellations (Figure 4). Flights on one day are cancelled together, so the day is the unit. Daily rates are very skewed (median 0.73%, maximum 57%) and their spread differs by season (Levene $W = 3.1$, $p = 0.026$), so a rank test.

Also considered	Why we did not use it
Chi-square test of independence (4×2)	counts are large enough (smallest expected 1,963), but it treats 336,776 flights as independent. Result: $\chi^2 = 1{,}055.3$ (3); its CI for winter − autumn (Newcombe, 2.26 to 2.55 pp) is 12 times narrower than the day-clustered one.
G-test, Fisher, logistic regression	same independence assumption as chi-square ($G = 1{,}173$)
ANOVA on daily rates	daily means are driven by two storm days. Welch form: $F = 7.26$ (3, 190), $p < 0.001$.

c) Result

Estimate	rates: winter 3.50%, summer 2.80%, spring 2.43%, autumn 1.10%; winter − autumn $= 2.40$ pp
95% confidence interval	0.69 to 4.12 pp (day-clustered)
Test statistic (d.f.)	$H = 39.9$ (3)
p-value	< 0.001
Decision	reject H_0

Seasons differ (a day-clustered Wald test on the flight-level rates agrees: $F = 6.99$ (3, 364), $p < 0.001$). But the interval for the largest gap reaches down to 0.69, below the 1-point threshold. Pairwise day-clustered tests (Holm) separate only autumn from each of the other three; winter, spring and summer cannot be told apart.

Winter's lead comes from one storm: 472 flights were cancelled on 8 February and 393 on 9 February. Dropping those two days puts winter (2.47%) below summer (2.80%). Dropping the 10 worst days, the test still rejects ($H = 39.7$, $p < 0.001$), with summer 2.69%, winter 1.92%, spring 1.82% and autumn 0.88%; winter − autumn is then 1.04 pp (CI 0.43 to 1.64). On a typical day (median daily rate) the gap is 0.67 pp (winter 1.08%, autumn 0.41%). Autumn is the lowest season in every version; that any season differs by more than 1 point is not shown.

Test 5 • Weekly delay patterns (P5): weekend versus weekday arrival delay

a) Hypotheses, written before running the test

H_0	$\mu_{\text{we}} - \mu_{\text{wd}} = 0$
H_1	$\mu_{\text{we}} - \mu_{\text{wd}} \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	Welch t-test on daily mean arrival delay (104 weekend days, 261 weekdays)
Smallest difference that matters	5 minutes

μ_{we}, μ_{wd}: mean daily arrival delay on Saturday–Sunday and Monday–Friday. Two-sided, because fewer flights (less congestion) and more leisure travellers (slower boarding) push in opposite directions. Weekend is a property of a day, so the day is the unit.

b) Other tests we considered

Two independent groups of days, unknown σ, and unequal spreads of the daily means (SD 13.4 vs 18.2 min; Levene $W = 6.0$, $p = 0.015$), so the Welch form.

Also considered	Why we did not use it
Welch t-test on flights	treats flights on one day as independent: $t = -39.1$ (160,784), CI -6.9 to -6.3. Clustered by day: -6.6 min (CI -9.9 to -3.3).
Pooled t-test	equal variances are rejected; $t = -3.66$ on days
Mann–Whitney U	compares ranks, not minutes. On daily means $p < 0.001$; on flights $P(\text{we} > \text{wd}) = 0.45$, medians -8 vs -4 min.
Paired t-test	a day belongs to one group only

c) Result

Estimate	-7.2 min (1.85 weekend, 9.06 weekday)
95% confidence interval	-10.6 to -3.8 min
Test statistic (d.f.)	$t = -4.15$ (254.8)
p-value	< 0.001
Decision	reject H_0

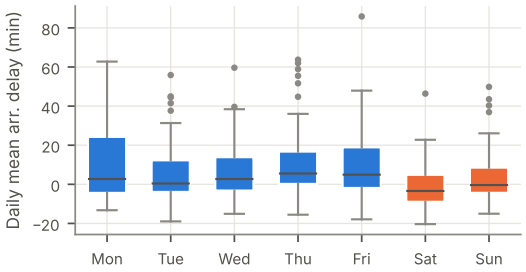


Figure 6: Daily mean arrival delay by day of the week. Saturday is lowest; Sunday looks like a weekday.

Weekend days have lower arrival delay. Whether the gap is more than 5 minutes is uncertain, because the interval reaches -3.8. Most of it is Saturday, the day with the fewest flights (38,720 scheduled, against about 50,000 on each weekday): Saturday flights average -1.45 min, 10.0 min less than weekday flights (CI -13.6 to -6.4, day-clustered). Sunday averages 4.82, 3.75 less (CI -8.1 to $+0.6$), which cannot be told apart from a weekday. Thursday is the worst day at 11.74 min.

Summary of the five tests

#	Estimate (95% CI)	Decision	Matters?
1	gain $+3.8$ min (2.7, 4.9)	fail to reject	no: mean below 5 min (median 7); padding, not catch-up
2	wet $+19.6$ min (14.8, 24.3)	reject ($p = 0.030$)	borderline: $p = 0.059$ after calendar adjustment
3	FL − AA $+19.8$ min (16.6, 22.9)	reject	only pairs with EV or FL; $\varepsilon^2 = 0.025$
4	winter − autumn $+2.4$ pp (0.7, 4.1)	reject	not shown above 1 pp; only autumn stands apart
5	weekend -7.2 min $(-10.6, -3.8)$	reject	real; above 5 min uncertain; a Saturday effect

With five tests, a Bonferroni level of $0.05/5 = 0.01$ changes one decision: Test 2 would not reject. With 300,000 rows almost every flight-level test rejects, so the useful columns are whether the interval clears the threshold and whether the result holds with the day as the unit.

6. Limitations

- **Association, not cause.** Precipitation comes with low visibility, wind and congestion; Test 2 cannot separate them, and its fixed-effects check removes only calendar and airport effects.
- **Dependence.** Clustering by day handles flights that share a day, but consecutive days are also correlated (lag-1 $r = 0.37$ for daily mean delay), so the day-level p-values are still somewhat optimistic. The Kruskal–Wallis p-value in Test 3 is flight-level; the Friedman and clustered checks cover it.
- **All flights, not a sample.** The data are every 2013 NYC departure, so p-values measure how unusual a pattern is against chance, not how well it carries to other years.
- **Survivor bias.** Delay tests use flights that flew. The worst hours cancel flights instead of delaying them, so Test 2 likely understates the effect of bad weather.
- **Weather at the origin, at the scheduled hour only.** Storms at the destination or a few hours earlier are not measured.
- **Choices we made.** The 5, 10 and 15-minute and 1-point thresholds follow from the FAA's 15-minute rule and the base rate; others would change the Matters column. Winter is January, February and December of 2013, so it joins two different winters.
- **Scope.** One year, one city, domestic flights. AirTran has since merged into Southwest and US Airways into American, so Test 3 is a 2013 snapshot.